

Akylas Stratigakos

UCL Energy Institute, University College London

a.stratigakos@ucl.ac.uk | <https://akylasstrat.github.io> | [Google Scholar](#) | [ORCID](#) | [GitHub](#)

Research Interests

My research develops machine-learning and optimization methods for reliable, low-carbon energy systems. I focus on decision-aware forecasting, robust and prescriptive analytics, and data-driven tools for power-system and building operations.

Academic Appointments

2025–present	Lecturer (Assistant Professor) <i>UCL Energy Institute, University College London, UK</i> Energy Systems and Data Analytics
2023–2025	Research Associate <i>EEE Department, Imperial College London, UK</i>

Education

2023	Ph.D., Energy and Processes <i>Mines Paris, PSL University, Sophia Antipolis, France</i> <i>Thesis:</i> Towards the Prescriptive Analytics Paradigm for Energy Forecasting and Power System Optimization
2016	MEng, Electrical and Computer Engineering <i>University of Patras, Patras, Greece</i> Specialization: Power Systems, Control

Visiting Positions

2022–2023	Visiting Researcher <i>OASYS Research Group, University of Málaga, Spain</i>
------------------	--

Publications

Journal Articles

- [1] J. G. De la Varga, S. Pineda, **A. Stratigakos**, and J. M. Morales. “End-to-End Pseudo-Measurement Learning for State Estimation under Limited Observability”. In: *Electric Power Systems Research* 265 (2027), p. 114220. [DOI](#).
- [2] G. Van Caelenberg, **A. Stratigakos**, and E. Spyrou. “Constructing Deployment Scenarios for Reserve Deliverability via Adaptive Robust Optimization”. In: *Electric Power Systems Research* 263 (2027). Presented at PSCC 2026, Limassol, Cyprus, 8–12 June 2026, p. 113632. [DOI](#).
- [3] **A. Stratigakos** and P. Andrianesis. “Learning Data-Driven Uncertainty Set Partitions for Robust and Adaptive Energy Forecasting with Missing Data”. In: *IEEE Transactions on Smart Grid* 17.1 (2026), pp. 806–817. [DOI](#).
- [4] C. Bergmeir, F. De Nijs, E. Genov, A. Sriramulu, M. Abolghasemi, R. Bean, J. Betts, Q. Bui, N. T. Dinh, N. Einecke, R. Esmailbeigi, S. Ferraro, P. Galketiya, R. Glasgow, R. Godahewa, Y. Kang, S. Limmer, L. Magdalena, P. Montero-Manso, D. Peralta, Y. P. S. Kumar, A. Rosales-Pérez, J. Ruddick, **A. Stratigakos**, P. Stuckey, G. Tack, I. Triguero, and R. Yuan. “Predict+Optimize Problem in Renewable Energy Scheduling.” In: *IEEE Access* 13 (2025), pp. 60064–60087. [DOI](#).

- [5] **A. Stratigakos**, S. Pineda, and J. M. Morales. “Decision-focused linear pooling for probabilistic forecast combination”. In: *International Journal of Forecasting* 41.3 (2025), pp. 1112–1125. ISSN: 0169-2070. [DOI](#).
- [6] **A. Stratigakos**, P. Andrianesis, A. Michiorri, and G. Kariniotakis. “Towards Resilient Energy Forecasting: A Robust Optimization Approach”. In: *IEEE Transactions on Smart Grid* 15.1 (2024), pp. 874–885. [DOI](#).
- [7] **A. Stratigakos**, S. Pineda, J. M. Morales, and G. Kariniotakis. “Interpretable Machine Learning for DC Optimal Power Flow With Feasibility Guarantees”. In: *IEEE Transactions on Power Systems* 39.3 (2024), pp. 5126–5137. [DOI](#).
- [8] K. Krommydas, C. Dikaiakos, G. Papaioannou, and **A. Stratigakos**. “Flexibility study of the Greek power system using a stochastic programming approach for estimating reserve requirements”. In: *Electric Power Systems Research* 213 (2022), p. 108620. [DOI](#).
- [9] K. F. Krommydas, **A. C. Stratigakos**, C. N. Dikaiakos, G. P. Papaioannou, M. G. Jones, and G. C. McLoughlin. “A Novel Modular Mobile Power Flow Controller for Real-Time Congestion Management Tested on a 150kV Transmission System”. In: *IEEE Access* 10 (2022), pp. 96414–96426. [DOI](#).
- [10] **A. Stratigakos**, S. Camal, A. Michiorri, and G. Kariniotakis. “Prescriptive trees for integrated forecasting and optimization applied in trading of renewable energy”. In: *IEEE Transactions on Power Systems* 37.6 (2022), pp. 4696–4708. [DOI](#).
- [11] **A. Stratigakos**, A. Bachoumis, V. Vita, and E. Zafiroopoulos. “Short-term net load forecasting with singular spectrum analysis and LSTM neural networks”. In: *Energies* 14.14 (2021), p. 4107. [DOI](#).
- [12] G. P. Papaioannou, C. Dikaiakos, **A. C. Stratigakos**, P. C. Papageorgiou, and K. F. Krommydas. “Testing the efficiency of electricity markets using a new composite measure based on nonlinear TS Tools”. In: *Energies* 12.4 (2019), p. 618. [DOI](#).

Conference Papers

- [13] K. F. Krommydas, **A. C. Stratigakos**, E. Chassioti, and I. Moraitis. “A two-stage stochastic unit-commitment formulation for evaluating the impact of battery energy storage systems on reserve requirements”. In: *IET Conference Proceedings CP904*. Vol. 2024. 29. IET. 2024, pp. 684–689. [DOI](#).
- [14] M. Kühnau, **A. Stratigakos**, S. Camal, S. Chevalier, and G. Kariniotakis. “Resilient Feature-driven Trading of Renewable Energy with Missing Data”. In: *2023 IEEE Power & Energy Society Innovative Smart Grid Technologies conference (ISGT)*. 2023, pp. 1–5. [DOI](#).
- [15] **A. Stratigakos**, D. van der Meer, S. Camal, and G. Kariniotakis. “End-to-end Learning for Hierarchical Forecasting of Renewable Energy Production with Missing Values”. In: *2022 17th International Conference on Probabilistic Methods Applied to Power Systems (PMAPS)*. 2022, pp. 1–6. [DOI](#).
- [16] **A. Stratigakos**, A. Michiorri, and G. Kariniotakis. “A Value-Oriented Price Forecasting Approach to Optimize Trading of Renewable Generation”. In: *2021 IEEE Madrid PowerTech*. 2021, pp. 1–6. [DOI](#).
- [17] K. F. Krommydas, **A. C. Stratigakos**, C. Dikaiakos, G. P. Papaioannou, E. Zafiroopoulos, and L. Ekonomou. “An improved flexibility metric based on kernel density estimators applied on the Greek power system”. In: *Flexitranstore*. Vol. 610. Springer, 2020, pp. 35–46. [DOI](#).
- [18] **A. C. Stratigakos**, K. F. Krommydas, P. C. Papageorgiou, C. Dikaiakos, and G. P. Papaioannou. “A Suitable Flexibility Assessment Approach for the Pre-Screening Phase of Power System Planning Applied on the Greek Power System”. In: *IEEE EUROCON 2019 -18th International Conference on Smart Technologies*. 2019, pp. 1–6. [DOI](#).

Preprints

- [19] **A. Stratigakos**, H. Wen, E. Spyrou, and P. Pinson. “Decision-calibrated prediction sets for robust power system operations”. In: (2026). working paper or preprint. arXiv: [2606.02081 \[math.OC\]](#). [Link](#).

Thesis

- [20] **A. Stratigakos**. “Towards the Prescriptive Analytics Paradigm for Energy Forecasting and Power System Optimization”. Theses. Université Paris sciences et lettres, July 2023. [Link](#).

Teaching Experience

2026–2027

Data Analytics for Sustainable Buildings (Co-Lead)

Sustainable Built Environments, Energy and Resources MEng (30 credits)

- Co-designing and co-leading the complete module for a cohort of 15 students.
- Teaching machine learning, time-series analysis, optimization, and control for smart-building operations.

- 2025–2027** **Energy Data Analytics in the Built Environment (Lead)**
Energy Systems and Data Analytics MSc (15 credits)
- Designed and currently lead a core MSc module for more than 60 students.
 - Teaching machine learning, time-series analysis, optimization, and control across built-environment applications.

Industry Experience

- 2022–2023** **Volunteer Data Scientist**
GIVMED, Athens, Greece
- Applied data analytics to redistribute surplus medical supplies to socially vulnerable groups and address healthcare inequalities.
- 2017–2020** **Research and Development Engineer**
Independent Power Transmission Operator (IPTO), Athens, Greece
- Developed tools to assess system flexibility under high renewable penetration.
 - Coordinated the successful installation of an innovative power flow controller.
 - Contributed to EU Horizon 2020 research and innovation projects on power-system flexibility.

Grants & Awards

- RESEARCH GRANTS **LLMs4Energy: Large Language Models for Energy Systems Modelling** —
UCL Energy Institute Early Career Research Fund (internal), Principal Investigator, £7.5k (2026)
- AWARDS Outstanding Reviewer IEEE Transactions on Sustainable Energy (2025)
Best PhD Thesis Award, Think Smartgrids Association (2024)
Erasmus+ Mobility Grant (2023)
Best Student Presentation & Travel Grant Award, 42nd International Symposium on Forecasting (2022)

Supervision

- PHD **Guillaume Van Caelenberg**, Imperial College London, 2023–2026 (expected).
Reserve Deliverability in Co-Optimized Electricity Markets (primary supervisor: Dr Elina Spyrou).
- Steven Sperazza**, University College London, 2025–2029 (expected). *Supporting policymaking with statistical emulation of complex models* (primary supervisor: Dr Erica Thompson).
- MASTER’S 8 MSc dissertations supervised
- UNDERGRADUATE 2 BSc/MEng dissertations supervised

Leadership and Service

REVIEWER	<i>Journals:</i> IEEE Transactions on Smart Grid; IEEE Transactions on Power Systems; IEEE Transactions on Sustainable Energy; Electric Power Systems Research. <i>Conferences:</i> IEEE PES PowerTech; International Conference on Probabilistic Methods Applied to Power Systems (PMAPS); Power Systems Computation Conference (PSCC).
ORGANIZATION	Workshop: Global Power System Transformation Consortium AI Workshop, at ICMS, Edinburgh, 19-23/02/2024 (<i>co-organizer, session chair</i>) Invited Session: Value-oriented Forecasting, 44th International Symposium on Forecasting, Dijon, France (<i>organizer, chair</i>). Seminar Series: Inaugural event of the CIGRE UK London Region Universities Hub, at Imperial College London, London, 06/05/2025 (<i>co-organizer</i>).
SERVICE	CIGRE UK University Hub London Region (<i>Vice Chair, 2025-2026</i>). 13th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys 2026) (<i>technical program committee member</i>)

Additional Information

LANGUAGES	Greek (native), English (proficient), German, French (basic)
SOFTWARE	Python, MATLAB, R, Julia
MEMBERSHIPS	Energy Institute, IEEE, CIGRE, ESIG, International Institute of Forecasters